



BIOMARKHER2 AI

AI-Enabled HER2 Biomarker Analysis Platform

INTRODUCTION

- Breast cancer is the most commonly diagnosed cancer among women worldwide and is increasing in India and Kerala. It arises from abnormal cell growth in breast tissue and varies widely in presentation and progression. Early detection through screening plays a crucial role in improving survival and treatment outcomes.
- Breast cancer is a heterogeneous disease, meaning that tumors differ in their biological characteristics, behavior, and response to treatment. Accurate classification is therefore essential and is primarily based on key biomarkers such as estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2). These biomarkers help determine the subtype of cancer, which directly influences prognosis and guides the selection of appropriate therapies.
- HER2 status is particularly important, as it influences both disease aggressiveness and eligibility for targeted therapies. It is assessed using **immunohistochemistry (IHC)** and categorized into four groups: 0, 1+, 2+, and 3+. Recent advances have highlighted the significance of **HER2-low tumors** (1+ and some 2+ cases), shifting interpretation from a simple positive/negative classification to a more nuanced spectrum. However, manual HER2 IHC scoring is subjective and prone to inter-observer variability, especially in borderline categories.
- This proposal introduces an **artificial intelligence (AI)-assisted system** to improve **accuracy and consistency** in **HER2 analysis** across **government medical colleges and cancer centres** in Kerala., with a focus on **HER2-low and equivocal cases**.

CURRENT SCENARIO

Current Diagnostic Tests Used in Breast Cancer

Breast cancer diagnosis and treatment planning rely on standardized biomarker testing. According to CAP guidelines, key tests include:

- **Estrogen Receptor (ER)** and **Progesterone Receptor (PR)** testing (hormone sensitivity)
- **Human Epidermal Growth Factor Receptor 2 (HER2)** testing (targeted therapy eligibility)

HER2 is initially assessed using **Immunohistochemistry (IHC)**, and equivocal cases are further confirmed using **In Situ Hybridization (ISH)** for gene amplification.

HER2 IHC Scoring System (CAP Guidelines)

HER2 expression is graded based on membrane staining intensity and pattern:

- **0** → No staining or faint/incomplete staining in <10% of tumor cells (**Negative**)
- **1+** → Faint/incomplete membrane staining in ≥10% of cells (**HER2-low**)
- **2+** → Weak to moderate complete membrane staining (**Equivocal**, requires ISH testing)
- **3+** → Strong complete membrane staining in ≥10% of cells (**Positive**)

Distinguishing between **0 vs 1+** and **1+ vs 2+** is particularly challenging and prone to observer variability.

CURRENT STATISTICS

Data Collected from	Kottayam Medical College
Sample Size	84
Duration	2024-25

HER2	COUNT
0	22
1	10
2	6
3	26

EXISTING SYSTEM

Currently, HER2 status in breast cancer is assessed manually by pathologists following guidelines from the College of American Pathologists (CAP). Tissue samples are stained using **immunohistochemistry (IHC)** and examined under a microscope to evaluate membrane staining intensity and pattern.

Step 1: Tissue Collection and Fixation

- Breast tumor tissue is obtained via biopsy or surgical excision
- Specimen is fixed in 10% neutral buffered formalin
- Inadequate fixation can compromise staining quality and interpretation

Step 2: Tissue Processing and Slide Preparation

- Fixed tissue is processed, embedded in paraffin, and sectioned
- Thin tissue sections are mounted on glass slides

Step 3: HER2 Immunohistochemistry (IHC)

- Slides are stained using HER2-specific antibodies
- Staining highlights HER2 protein on tumour cell membranes
- Results depend on antibody clone, reagent quality, and lab protocols

EXISTING SYSTEM

Step 4: Manual Microscopic Evaluation

- Pathologists visually assess staining intensity and completeness
- HER2 score (0, 1+, 2+, or 3+) is assigned based on guidelines

Step 5: Reflex FISH/ISH Testing for IHC 2+

- Equivocal cases undergo FISH/ISH to evaluate gene amplification
- HER2/CEP17 ratios are calculated
- Results determine final HER2 status

Step 6: Final Reporting

- IHC and FISH results are integrated
- Final HER2 status guides oncologist treatment decisions

The FISH/ISH Testing process relies heavily on the expertise and experience of the pathologist.

However, the current system has several limitations. HER2 scoring, especially in 1+ and 2+ categories, is subjective and prone to inter- and intra-observer variability. Differences in interpretation, staining quality, and workload can affect consistency, leading to potential variations in diagnosis and treatment decisions.

PROBLEM WITH THE CURRENT PROCESS

The current HER2 assessment process, based on manual evaluation of immunohistochemistry (IHC) slides is highly dependent on visual interpretation and pathologist expertise. Despite standardized guidelines, several challenges exist, including:

- Differentiating 1+ from background staining and 2+ from 3+ is visually subjective, leading to significant inter- and intra-observer variability.
- HER2 expression can vary across different tumor regions, and small biopsy samples may not represent the entire tumor, affecting scoring reliability.
- Variations in tissue fixation, antibody clones, staining platforms, and laboratory protocols can result in inconsistent results across centers.
- Accurate distinction between 0 and 1+ is difficult but clinically critical; misclassification may exclude patients from emerging targeted therapies.
- Equivocal (2+) cases require confirmatory in situ hybridization (ISH) testing, which is costly, time-consuming, and not uniformly accessible.
- Sequential testing (IHC followed by ISH) increases turnaround time, potentially delaying treatment decisions and initiation of targeted therapy.

NEED OF THE PROJECT

- Accurate assessment of human epidermal growth factor receptor 2 (HER2) status is critical for breast cancer treatment planning, particularly with the growing clinical importance of HER2-low (1+) and equivocal (2+) cases. The current process, based on manual interpretation of immunohistochemistry (IHC) slides is time-consuming and highly dependent on pathologist expertise.
- HER2 scoring in borderline categories is subjective and prone to inter- and intra-observer variability, where small differences in interpretation can significantly impact treatment eligibility. Additionally, reliance on reflex in situ hybridization (ISH/FISH) testing increases cost and turnaround time, particularly in public healthcare settings with high workload.
- To address these challenges, an **AI-assisted system for HER2 IHC scoring** is proposed to provide objective, standardized, and reproducible results across government medical colleges and cancer centres in Kerala. This system will support pathologists in improving diagnostic accuracy, reducing variability and bias, optimizing the use of confirmatory testing, and shortening reporting time, thereby enabling more consistent and timely treatment decisions.

PROJECT OBJECTIVE- APPROACH 1

To develop an AI-enabled HER2 biomarker analysis platform for accurate and consistent HER2 IHC classification, with a focus on borderline (2+) and HER2-low (1+) cases to support improved clinical decision-making.

Specific Objectives:

1. Assist pathologists in accurate and consistent HER2 IHC classification, especially in borderline cases (0/1+ vs 2+, 2+ vs 3+).
2. Reduce inter- and intra-observer variability through standardized, quantitative analysis of staining patterns.
3. Reduce dependency on costly confirmatory ISH/FISH testing, particularly in equivocal (2+) cases.
4. Ensure that patients receive appropriate targeted therapies based on accurate HER2 biomarker classification.
5. Decrease diagnostic turnaround time and reduce the workload of pathologists by automating HER2 analysis.

PROJECT OBJECTIVE - APPROACH 2

1. To develop an AI-assisted system for **quantitative analysis of color intensity** in digital IHC slides, for HER2 staining in WSI.
2. To convert **subjective visual interpretation** of staining into **objective, measurable outputs**.
3. To design algorithms for **tissue detection, stain separation, and intensity quantification** at patch/region level.
4. To enable **classification of staining intensity** into standardized categories such as weak, moderate, and strong.
5. To compute **area-based distribution of intensity**, providing percentage of tissue under each intensity level.
6. To generate **spatial heatmaps** for visual representation of staining patterns and heterogeneity across the slide.
7. To provide **structured quantitative reports** that support pathologists in decision-making.
8. To reduce **inter-observer variability** and improve consistency in HER2 interpretation.
9. To build a **scalable AI foundation** that can be extended to bio marker for full HER2 scoring analysis in future phases.

PROJECT SCOPE

1. Processing of Automated IHC HER 2 positive slides and Digital processing of the IHC slides into WSI
2. Implementation of tissue detection to exclude background regions.
3. Application of stain normalization and color deconvolution to isolate DAB intensity.
4. Patch extraction and patch-level analysis for handling large image sizes efficiently.
5. Development of AI/ML models for:
 - Intensity classification (weak / moderate / strong)
 - Or continuous intensity estimation
6. Calculation of:
 - Total tissue area
 - Area percentage under each intensity category
7. Generation of:
 - Heatmaps (intensity distribution visualization)
 - Slide-level quantitative summary reports
8. Basic validation against pathologist annotations using statistical metrics (accuracy, agreement). Kappa..
9. Design for integration into existing digital pathology workflow as a **pre-scoring tool**